Trigonometry Exact Values

Exam code: 9MA0

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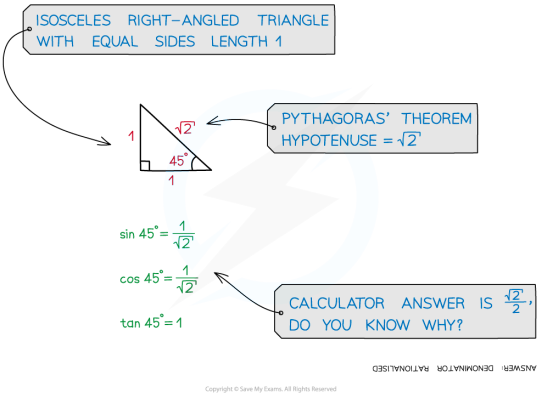
Trigonometry exact values

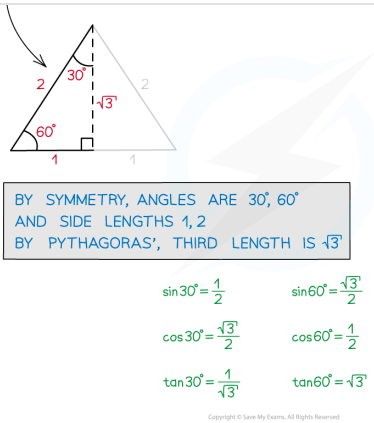
What are exact values in trigonometry?

- In trigonometry some values of sin, cos and tan are "nice"
- For example **sin 30° = ½, tan 45° = 1**
- We can find those associated with the angles **30°**, **45°** and **60°** using SOHCAHTOA on isosceles and equilateral triangles



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



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- **sin**, **cos** and **tan** values for **30°**, **45°** and **60°** should be recalled easily
- Memorise either the actual values or how to work them out from the triangle
- **sin**, **cos** and **tan** for **0°**, **90°**, **180°** should also be very familiar
- They can be recalled using the relevant **trigonometric graph** (see Graphs of Trigonometric Functions)

What exact values do I need to know in trigonometry?

- All of the above applies to radians as well as degrees
- Here is a table of all exact values including **radians** (see Radian Measure)

DEGREES	0°	30°	45°	60°	90°	180°	360°
RADIANS	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	−1	1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	UNDEFINED	0	0

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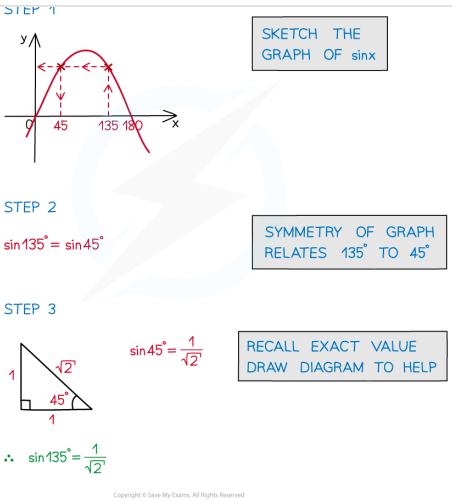
How do I find exact values of other angles?

- Exact **sin**, **cos** and **tan** values of **multiples** of **30°**, **45°**, **60°** can also be found
- This is a combination of recalling the basic values and using the graph



Examiner Tips and Tricks

- Draw the triangles for **sin**, **cos** and **tan** of **30°**, **45°** and **60°** on the exam paper so that you can then use them as many times as you need throughout the paper.

Worked Example

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